

Software Requirements Specification (SRS)

SmartFarm Management Application

Document Version: 1.0

SDLC Stage: Phase 1 — Requirements Status: Approved Baseline Specification

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document details the complete functional, non-functional, security, and architectural requirements for the **SmartFarm Management Application**. It serves as the authoritative single source of truth for engineering, testing, and continuous maintenance.

1.2 Scope of the System

SmartFarm is an enterprise-grade, zero-cost cloud farm management platform designed for agricultural enterprises. The application enables:

Executive oversight and financial tracking by **Farm Administrators**.

Sector-level resource management, project planning, and supervisor delegation by **Farm Managers**.

Field-level yield, activity, and input tracking by **Field Supervisors** under strict financial privacy isolation.

1.3 Definitions, Acronyms, and Abbreviations

RBAC: Role-Based Access Control.

PBAC: Privilege/Policy-Based Access Control (fine-grained feature toggle permissions).

1:N: One-to-Many entity relationship.

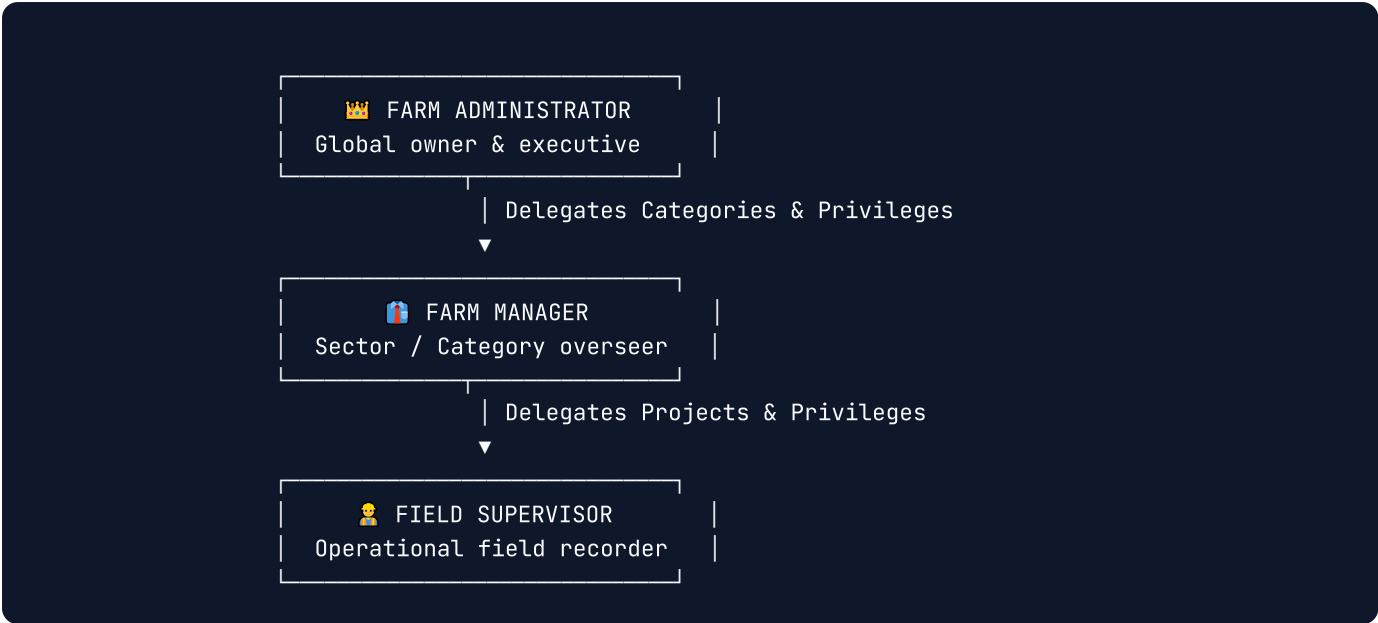
Zero-State Dashboard: A tailored, clean UI rendered when an authenticated user has zero assigned projects/categories, suppressing empty graphs and displaying actionable next steps.

KPI: Key Performance Indicator (Revenue, Yield, Expenses, Net Profit).

2. Overall Description

2.1 Product Perspective & User Personas

The system enforces a strict 3-tier organizational hierarchy:



Persona	Role Key	System Scope	Max Quota	Core Responsibilities
Farm Administrator	`ADMIN`	Entire Farm (Global)	1 Account	User management, financial health, category assignment, manager privilege delegation, system configuration.
Farm Manager	`MANAGER`	Assigned Categories	Max 2 Accounts	Project planning, supervisor hiring & assignment, sector expense auditing, supervisor privilege delegation.
Field Supervisor	`SUPERVISOR`	Assigned Projects	Max 10 Accounts	Daily field logging (harvests, activities, stock consumption). **Zero access to sales revenue/budgets.**

2.2 General Constraints

Dedicated Supervisor Model: Supervisors are **not shared** across managers. A supervisor belongs to exactly one Manager (`manager_id`).

One-to-Many Project Assignment: A supervisor can oversee multiple projects, but each project is assigned to at most one supervisor.

Financial Privacy Barrier: Field supervisors are strictly prohibited from viewing total farm sales cash flow, project budgets, profit margins, or net values.

Zero-Cost Cloud Infrastructure: Designed to run in production with high availability at \$0/month (Vercel Frontend, Render Backend, Cloud MySQL/TiDB, Gmail SMTP).

3. Detailed Functional Requirements (FR)

FR-1: Authentication, Account Provisioning & Password Recovery

FR-1.1: System shall support authentication using either `Username` or `Email` combined with a password.

FR-1.2: Unauthenticated users can request a password reset by providing their registered email address (`POST /users/forgot-password`).

FR-1.3: If an email is not registered in the system, the API shall return HTTP 400 with: `"No account found with this email address. If you did not register an email, please contact your Farm Administrator to reset your password."`

FR-1.4: If the email exists, the system generates a secure cryptographic token (`PasswordResetToken` valid for 1 hour) and sends an HTML reset email via SMTP.

FR-1.5: Administrators and Managers can execute direct password resets for staff accounts via the User Management portal (`PATCH /users/{id}/admin-reset-password`).

FR-2: Staff Quotas & Hierarchical Access Delegation (PBAC)

FR-2.1: The system shall strictly enforce organizational staff limits: Max 1 Admin, Max 2 Managers, Max 10 Supervisors.

FR-2.2: Admin to Manager Delegation:

- Admin assigns one or more Farm Categories to a Manager.
- Admin configures fine-grained privilege toggles for the Manager:
 - `CAN_CREATE_CATEGORIES` : Ability to create new farm sectors.
 - `CAN_CREATE_SUPERVISORS` : Ability to provision supervisor accounts.
 - `CAN_VIEW_FINANCIALS` : Ability to view profit/loss and revenue totals.
 - `CAN_MANAGE_BUDGETS` : Ability to set and modify project budgets.

FR-2.3: Manager to Supervisor Delegation:

- Manager provisions supervisors linked to their manager account ID (`manager_id`).
- Manager assigns specific projects (within the Manager's categories) to the Supervisor.
- Manager configures fine-grained privilege toggles for the Supervisor:
 - `CAN_RECORD_HARVEST` : Log harvest yield volumes and units.
 - `CAN_LOG_ACTIVITIES` : Submit field activity and labor logs.
 - `CAN_USE_INVENTORY` : Deduct fertilizer, feed, and tools from stock.
 - `CAN_RECORD_EXPENSES` : Log petty field cash expenses.
 - `CAN_RECORD_SALES` : Record immediate farm-gate sales without viewing global finances.

FR-3: Workload & Capacity Validation Rules

FR-3.1 (Manager Workload): When assigning a category to a Manager, the system shall evaluate existing assignments. If a manager exceeds 3 active categories, an advisory workload confirmation is triggered.

FR-3.2 (Supervisor Workload): Each supervisor has a `max_project_capacity` (default 4 projects). The system blocks assigning additional active projects unless capacity is adjusted or existing projects are completed.

FR-4: Category & Project Lifecycle Management

FR-4.1: Projects must belong to an existing Category and possess a globally unique project name.

FR-4.2 (Date Validation Rule): The system shall strictly validate that `startDate ≤ endDate`. If `startDate > endDate`, both backend and frontend reject the request with: `"Start date cannot be greater than end date!"`

FR-4.3 (Unique ID Generation): Project, Category, and Transaction IDs are generated using an incremental collision-safe loop (`while (repo.existsById(id)) count++;`) preventing primary key collisions after deletions.

FR-5: Operational Field Tracking (Harvest, Activity, Inventory)

FR-5.1: Harvest records capture item name, numerical yield, unit of measurement (KG, Litres, Bags), date, and notes.

FR-5.2: Activities capture activity title, type (Weeding, Irrigation, Spraying, Feeding), date, and notes.

FR-5.3: Inventory tracks quantity in stock, unit prices, reorder levels, and records deduction history when consumed by projects.

FR-6: Financial Governance & Sales Enforcement

FR-6.1 (Stock-Backed Sales Rule): A project cannot record a sale exceeding its total recorded harvest yields minus previous sales.

FR-6.2 (Supervisor Financial Shield): Supervisor API endpoints and dashboard views strip budget allocations, sales revenue, and profit calculations.

FR-7: Dynamic Scoped Dashboards & Zero-State Engine

FR-7.1 (Admin Dashboard): Renders enterprise-wide aggregates, category financial distribution, user quotas, and global cashflow.

FR-7.2 (Manager Dashboard): Aggregates budget, project counts, and supervisor workloads strictly for the manager's assigned categories.

FR-7.3 (Supervisor Dashboard): Aggregates harvest volumes, pending tasks, and recent logs strictly for assigned projects.

FR-7.4 (Zero-State Suppression): If a user has 0 assigned categories or 0 assigned projects:

- All metrics cards, financial figures, and graphs are **suppressed**.
- Renders a dedicated **Action Hero Card** providing immediate contextual instructions (e.g., `"No categories assigned yet. Contact your Farm Administrator to get started."`).

FR-8: User Deletion & Referential Integrity Safeguards

FR-8.1: Admin accounts cannot be deleted (`isUserAdmin` protection).

FR-8.2: Deleting a user safely cascades/unlinks child records: deletes active `PasswordResetToken` rows, clears category join tables, unlinks `supervisor_id` on projects, and clears `createdById` references.

4. Non-Functional Requirements (NFR)

NFR-1 (Security): BCrypt password hashing (\$cost = 10\$), role-based endpoint security, parameter sanitization, and SQL injection prevention via Spring Data JPA.

NFR-2 (Performance): Page load times $\leq 1.5s$ on broadband; database indexed on `username`, `email`, `category_id`, `supervisor_id`, `manager_id`.

NFR-3 (Reliability): 99.9% uptime target backed by automated heartbeat monitoring (UptimeRobot).

NFR-4 (Cost): 100% compliant with zero-cost cloud architecture limits (Render Free Web Service, Vercel Hobby, TiDB Cloud / Aiven Free Tier).

NFR-5 (Usability): Typography optimized with `Inter` and `DM Sans`, accessible contrast ratios, and responsive mobile-first views.